

Part VIII — Electricity and Magnetism · Chapter 41

Electric Fields and Electric Potential

How charges create fields in space, and how potential energy per charge describes the electrical landscape

A compass needle far from any magnet points steadily north. Remove the magnet from the room and the compass still responds — it is responding to a magnetic field that permeates the space even in the magnet's absence. Similarly, a test charge placed near another charge feels a force even though the two charges never touch. The ancient puzzle of "action at a distance" — how can one object push on another across empty space? — is resolved by the field concept: each charge alters the properties of the space around it, creating an electric field, and that field acts locally on any other charge placed in it. The field model is not just a mathematical convenience; it is the correct description of nature. Electromagnetic fields carry energy, travel at the speed of light, and exist independently of the charges that created them. This chapter introduces the electric field, the electric potential, and the powerful analytical framework they provide.

41.1 Why Physicists Use Field Models

Before the field concept, electric and gravitational forces were described as "action at a distance": one body exerted force directly on another with nothing between them. Newton himself was uncomfortable with this idea but could not explain how gravity propagated across empty space.

Michael Faraday (1791–1867), who had no formal mathematical training but extraordinary physical intuition, introduced the field concept. He imagined that a charged object modifies the space around it in a way that is physically real and that other charges respond to this modification — not to the distant charge directly. The field exists at every point in space and carries both energy and momentum.

The field concept is not merely a bookkeeping device. It resolves the action-at-a-distance problem by specifying that charge A creates a field that travels outward at the speed of light, and that field (not charge A directly) exerts force on charge B. If charge A is suddenly moved, the field change propagates at the speed of light, and charge B "finds out" about the change only after time r/c has elapsed. This is physically correct:

nothing in electromagnetism acts instantaneously over distance.

The field concept is also mathematically powerful. Instead of tracking every pairwise interaction between charges, we can (1) compute the field created by one charge distribution, then (2) find the force on any other charge by multiplying charge $\times$ field. The two steps are independent, which greatly simplifies complex problems.

Historical Note: Michael Faraday (1791–1867) came from a poor background and received almost no formal schooling, yet became one of the greatest experimental physicists ever. He introduced the concepts of electric and magnetic fields, discovered electromagnetic induction (the basis of all electric generators), invented the electric motor concept, and made fundamental contributions to electrochemistry. James Clerk Maxwell later provided the mathematical framework for Faraday's field ideas in Maxwell's equations (1865).

41.2 What an Electric Field Is

The electric field E at a point in space is defined as the electric force per unit positive test charge that would be experienced by a small positive test charge placed at that point:

$$\mathbf{E} = \mathbf{F} / q$$

Electric field definition. E = electric field (N/C or V/m); F = electric force (N) on a test charge; q = test charge (C), taken as positive. The electric field is a vector: it has both magnitude and direction at every point in space.

The test charge is assumed to be infinitesimally small so that it does not disturb the field it is measuring. In practice, "small positive test charge" means a charge small enough that its presence does not significantly alter the charge distribution creating the field.

The electric field is a vector field: at every point in space there is a vector (an arrow) specifying the strength and direction of the electric influence at that location. The SI unit of electric field is newtons per coulomb (N/C), or equivalently, volts per metre (V/m) — two names for the same unit, as we will show in Section 41.12.

Once the electric field at a point is known, the force on any charge q placed at that point is:

$$\mathbf{F} = q\mathbf{E}$$

Force on a charge in an electric field. F (N) = charge q (C) $\times$ field E (N/C). For positive q , force is in the direction of E ; for negative q , force is opposite to E .

41.3 Field Direction and Field Lines

The direction of the electric field at a point is the direction of the force that would be exerted on a small positive test charge placed at that point. Since like charges repel, the field of a positive charge points away from the charge (a positive test charge would be pushed away). The field of a negative charge points toward the charge (a positive test charge would be attracted inward).

Electric field lines (or lines of force) are a visual representation of the electric field. They are drawn so that:

- The direction of the field line at any point is tangent to the line at that point.
- The density of lines (number of lines per unit area crossing a surface perpendicular to the lines) is proportional to the magnitude of the field.
- Field lines begin on positive charges and end on negative charges (or travel to/from infinity).
- Field lines never cross each other. (If they did, the field would have two directions at that point, which is impossible.)

For an isolated positive point charge, field lines radiate outward uniformly in all directions. For an isolated negative charge, they converge inward. For two opposite charges (a dipole), lines leave the positive charge and curve to arrive at the negative charge. For two like charges, the lines bend away from the region between the charges and no lines cross the midpoint plane.

Key Point: Field lines are not paths that charges travel. They show the direction a positive charge would accelerate at each point, not the trajectory it would follow (which depends on its initial velocity and on all subsequent forces).

41.4 Electric Field Around Point Charges

Using Coulomb's Law to find the force on a test charge q_0 due to a source charge Q , and then dividing by q_0 , gives the electric field due to a point charge:

$$E = k |Q| / r^2$$

Electric field magnitude due to a point charge Q at distance r . E (N/C); $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$; Q = source charge (C); r = distance from Q to the field point (m). The field is radially outward from Q if Q is positive; radially inward if Q is negative.

The field due to a point charge is spherically symmetric: at any fixed distance r from Q , the field has the same magnitude in all directions. The field diminishes as $1/r^2$, just as the Coulomb force does. This is expected: $E = F/q$ and both F and E depend on r in the same way.

Example — Field of a single charge Find the electric field at a distance of 0.25 m from a charge of $+5.0 \mu\text{C}$. $E = kQ/r^2 = (8.99 \times 10^9)(5.0 \times 10^{-6})/(0.25)^2 = (4.495 \times 10^5)/(0.0625) = 7.2 \times 10^6 \text{ N/C} = 7.2 \text{ MN/C}$, directed away from the charge.

Example — Force from field An electron ($q = -1.602 \times 10^{-19} \text{ C}$) is placed where the electric field is $E = 4.0 \times 10^4 \text{ N/C}$ directed in the $+x$ direction. Find the force. $F = qE = (-1.602 \times 10^{-19})(4.0 \times 10^4) = -6.4 \times 10^{-15} \text{ N}$. The negative sign means the force is in the $-x$ direction — opposite to E , because the electron has negative charge.

41.5 Superposition of Electric Fields

The total electric field at a point due to multiple charges is the vector sum of the fields due to each individual charge. This is the superposition principle, applied to fields rather than forces.

$$\mathbf{E}_{\text{total}} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \dots$$

Superposition of electric fields. Each E_i is computed from the corresponding source charge using $E = kQ/r^2$ with the appropriate direction. The vectors are then added component by component.

The procedure is:

Step 1: Identify: All source charges and the field point.

Step 2: Compute each field: Find the magnitude $E_i = k|Q_i|/r_i^2$ for each source charge and determine the direction (away from positive charges, toward negative charges).

Step 3: Resolve into components: Express each field vector as x and y components using geometry.

Step 4: Add components: $E_x = \sum E_{ix}$; $E_y = \sum E_{iy}$.

Step 5: Find magnitude and direction: $E = \sqrt{E_x^2 + E_y^2}$; $\theta = \arctan(E_y/E_x)$.

Example — Superposition: two charges Two charges: $q_1 = +3.0 \mu\text{C}$ at origin, $q_2 = -3.0 \mu\text{C}$ at $(0.40 \text{ m}, 0)$. Find the field at $P = (0.20 \text{ m}, 0)$ (midpoint). From q_1 : $E_1 = k(3 \times 10^{-6})/(0.20)^2 = (8.99 \times 10^9)(3 \times 10^{-6})/0.04 = 6.74 \times 10^5 \text{ N/C}$, pointing in $+x$ direction (away from positive charge). From q_2 : $r = 0.20 \text{ m}$, $E_2 =$ same magnitude $= 6.74 \times 10^5 \text{ N/C}$, pointing in $+x$ direction (toward negative charge). Total $E = 2 \times 6.74 \times 10^5 = 1.35 \times 10^6 \text{ N/C}$ in the $+x$ direction. Both fields point in the same direction at the midpoint of a dipole.

41.6 Electric Force from an Electric Field

The electric field provides a compact way to describe the effect of a charge distribution on any other charge placed in the field. Given the field E at a location, the force on a charge q placed there is simply:

$$\mathbf{F} = q\mathbf{E}$$

Force from field. This works for any charge q , positive or negative, large or small. Positive q gives force parallel to E ; negative q gives force antiparallel to E .

This relationship is why the field is defined per unit positive charge: it gives the force per unit charge, so the actual force on any charge q is always $F = qE$, regardless of the sign or magnitude of q .

For a charged particle of mass m in an electric field, Newton's second law gives its acceleration:

$$\mathbf{a} = \mathbf{F}/m = q\mathbf{E}/m$$

Acceleration of a charged particle in a field. The particle may also have gravitational acceleration g acting, which must be added vectorially if relevant.

This relationship is exploited in particle accelerators (protons and electrons are accelerated in electric fields), cathode ray tubes (electrons are deflected by electric fields to steer the beam), mass spectrometers (ions are separated by the ratio of charge to mass, q/m), and inkjet printers (charged droplets are steered by electric fields).

41.7 Uniform Electric Fields

A uniform electric field has the same magnitude and direction at every point in a region. The field lines are parallel and equally spaced. The most common way to create a uniform field is with two large, parallel, conducting plates carrying equal and opposite charges (a parallel-plate capacitor).

Between the plates of a parallel-plate capacitor separated by distance d with potential difference V (voltage), the uniform field has magnitude:

$$\mathbf{E} = \mathbf{V} / d$$

Electric field between parallel plates. E (V/m or N/C) = V (volts) $\div$ d (metres). The field points from the positive plate to the negative plate. Valid for the region well away from the plate edges.

In a uniform field, the force on a charged particle is constant (same magnitude and direction everywhere). This means a charged particle in a uniform field undergoes constant acceleration, analogous to projectile motion in a gravitational field. The kinematic equations from Chapter 3 apply directly with the electric acceleration $a =$

qE/m replacing g .

Example — Electron in a uniform field Two parallel plates are 5.0 mm apart with $V = 100$ V. $E = V/d = 100/0.005 = 20\,000$ V/m. An electron ($q = 1.602 \times 10^{-19}$ C, $m = 9.109 \times 10^{-31}$ kg) enters parallel to the plates. Acceleration: $a = qE/m = (1.602 \times 10^{-19})(20\,000)/(9.109 \times 10^{-31}) = 3.52 \times 10^{14}$ m/s² — enormous by everyday standards, but electrons have very small mass.

41.8 Electric Potential Energy

Just as gravity is a conservative force (meaning work done by gravity depends only on start and end positions, not on the path), the electric force is also conservative. This means we can define an electric potential energy (U) that depends on position.

When a charge q moves from point A to point B in an electric field, the work done by the electric field is:

$$W_{AB} = -\Delta U = -(U_B - U_A) = U_A - U_B$$

Work-energy relation for conservative forces. If the electric field does positive work on the charge (moving it in the direction of force), potential energy decreases. If the charge moves against the field, potential energy increases.

For two point charges Q and q separated by distance r , the electric potential energy is:

$$U = k Q q / r$$

Electric potential energy of two point charges. U (joules); $k = 8.99 \times 10^9$ N·m²/C²; Q, q = charges (C); r = separation (m). U is positive if the charges have the same sign (energy stored in repulsion); negative if opposite (energy released in attraction).

The reference point (where $U = 0$) is chosen at $r = \infty$ (infinite separation). This means two repelling charges have positive potential energy — they would accelerate away from each other, converting potential energy to kinetic energy. Two attracting charges have negative potential energy — energy must be supplied to separate them.

41.9 Electric Potential (Voltage)

Electric potential V (voltage) is the electric potential energy per unit charge. Like the electric field is force per charge, potential is energy per charge:

$$V = U / q$$

Electric potential (voltage) V (volts, V) = potential energy U (joules) $\div$ charge q (coulombs). 1 volt = 1 joule per coulomb ($1 V = 1 J/C$). Potential is a scalar — it has magnitude but no direction.

The electric potential due to a single point charge Q at distance r is:

$$V = k Q / r$$

Electric potential of a point charge. V (volts); $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$; Q = source charge (C); r = distance from Q (m). Positive Q gives positive V ; negative Q gives negative V .

Potential is a scalar, which makes it easier to work with than the vector electric field in many problems. The potential due to multiple charges is the algebraic (scalar) sum of the potentials due to each charge: $V_{\text{total}} = V_1 + V_2 + V_3 + \dots$. No vector addition is required.

The potential energy of a charge q at a location where the potential is V is $U = qV$. The work done moving a charge q through a potential difference ΔV is:

$$W = q \Delta V = q(V_B - V_A)$$

Work done by an external agent moving charge q from potential V_A to V_B . If the charge moves to higher potential ($V_B > V_A$), the external agent does positive work (or the field does negative work). Positive charges naturally move from high to low potential (like masses falling to lower height).

Example — Potential of a point charge Find the electric potential 0.30 m from a charge of $+6.0 \mu\text{C}$. $V = kQ/r = (8.99 \times 10^9)(6.0 \times 10^{-6})/0.30 = 1.80 \times 10^5 \text{ V} = 180 \text{ kV}$. Note: if we asked for the potential energy of a $2.0 \mu\text{C}$ charge placed at this point, $U = qV = (2.0 \times 10^{-6})(1.80 \times 10^5) = 0.36 \text{ J}$.

41.10 Potential Difference

Potential difference (voltage) between two points A and B is $\Delta V = V_B - V_A$. It is the work done per unit charge to move a positive test charge from A to B:

$$\Delta V = V_B - V_A = W_{\{A \rightarrow B\}} / q$$

Potential difference ΔV (volts). $W_{\{A \rightarrow B\}}$ is the work done by an external agent in moving charge q from A to B against the electric field. Unit: volt = joule/coulomb.

Potential difference is what voltmeters measure. A battery maintains a potential difference between its terminals: a 9 V battery keeps its positive terminal at 9 V higher potential than its negative terminal. This drives current through a circuit.

The electron-volt (eV) is a useful unit of energy in atomic and particle physics: 1 eV = the kinetic energy gained by one elementary charge accelerated through 1 V of potential difference = $(1.602 \times 10^{-19} \text{ C})(1 \text{ V}) = 1.602 \times 10^{-19} \text{ J}$. A proton accelerated through 1000 V gains 1000 eV = 1 keV of kinetic energy. This unit will be essential in nuclear and particle physics (Chapters 55–58).

Example — Potential difference and kinetic energy A proton ($q = 1.602 \times 10^{-19} \text{ C}$, $m = 1.673 \times 10^{-27} \text{ kg}$) starts from rest and is accelerated through a potential difference of 5000 V. Find its final kinetic energy and speed. $KE = q\Delta V = (1.602 \times 10^{-19})(5000) = 8.01 \times 10^{-16} \text{ J}$. $KE = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2KE/m} = \sqrt{(2 \times 8.01 \times 10^{-16} / 1.673 \times 10^{-27})} = \sqrt{(9.59 \times 10^{10})} \approx 9.8 \times 10^5 \text{ m/s} \approx 980 \text{ km/s}$. This is about 0.3% of the speed of light — non-relativistic, so our formula is valid.

41.11 Equipotential Surfaces

An equipotential surface (or equipotential line in two dimensions) is a surface (or curve) on which the electric potential is the same at every point. Moving a charge along an equipotential surface requires no work because $\Delta V = 0$ and $W = q\Delta V = 0$.

Equipotential surfaces and electric field lines are always perpendicular to each other. This follows because the electric field has no component along an equipotential surface (if it did, moving a charge along the surface would require work). The field is entirely perpendicular to the equipotential surface.

For a positive point charge, equipotential surfaces are concentric spheres centred on the charge. Field lines are radial lines crossing these spheres at right angles. For a uniform field between parallel plates, equipotential surfaces are planes parallel to the plates, equally spaced; field lines are perpendicular to the plates.

Conductors in electrostatic equilibrium are equipotential surfaces: all points on or inside a conductor (when no current flows) are at the same potential. This is why

charge distributes on the outer surface of a conductor and why the field inside a conductor in electrostatic equilibrium is zero. (Any interior field would drive currents, which would redistribute charge until the field vanishes.)

Key Point: The relationship between equipotentials and field lines is the electrostatic analogy of contour lines on a topographic map: contour lines (equal height = equal gravitational potential energy) are always perpendicular to the direction of steepest descent (the direction a ball would roll = the direction of gravitational force). Electric potential plays the role of height; field lines play the role of slope direction.

41.12 Relationship Between Field and Potential

The electric field and electric potential are two ways of describing the same physical situation. They are related: the field is the negative rate of change of potential with distance (the negative gradient of potential). In one dimension:

$$E_x = -\Delta V / \Delta x$$

Electric field from potential. E_x (V/m) = negative of the rate of change of potential in the x direction. The negative sign: field points from high potential to low potential, in the direction of decreasing V . Units: $V/m = N/C$ (both are valid for electric field).

This relationship shows that (a) the units V/m and N/C are equivalent (both are SI units of electric field), and (b) the field is stronger where the equipotentials are more closely spaced (steeper potential gradient), just as gravity is stronger where topographic contours are more closely spaced (steeper slope).

This relation also explains why the electric field inside a conductor in electrostatic equilibrium is zero: the conductor is an equipotential (no change in V), so $\Delta V/\Delta x = 0$, so $E = 0$. Conversely, this is why high fields are concentrated at sharp points and edges of conductors: the potential must change rapidly near a sharp tip, producing a large gradient and therefore a large field. Lightning rods exploit this: the very strong field at the tip of the rod ionises the surrounding air and provides a low-resistance discharge path for lightning.

Example — Field from potential A potential difference of 300 V exists between two parallel plates 6.0 mm apart. $E = \Delta V/d = 300/0.006 = 50\,000 \text{ V/m} = 50 \text{ kV/m}$, directed from the high-potential plate to the low-potential plate.

41.13 Applications of Electric Fields and Potential

The electric field and potential framework has applications across all of science and technology.

Cathode Ray Tubes (CRT)

The first electronic displays (oscilloscopes, early TVs) used cathode ray tubes: an electron gun accelerates electrons through a large potential difference, and deflecting plates (uniform fields) steer the beam. The electron enters the deflecting region at high speed and is deflected sideways by the electric field, tracing a path analogous to projectile motion but in an electric field rather than a gravitational one.

Electrocardiograms and Electroencephalograms

The electrical activity of the heart (ECG) and brain (EEG) creates potential differences measurable at the skin surface. The heart's depolarisation wave creates a dipole-like potential pattern that varies with time as the wave propagates through cardiac muscle. Electrodes on the skin measure these potential differences (typically millivolts), and the pattern allows diagnosis of cardiac rhythm, infarction, and conduction abnormalities.

Electrostatic Shielding

A conductor enclosing a region shields the interior from external electric fields. Because the conductor is an equipotential, any external field terminates on its outer surface; the interior field is zero. This is a Faraday cage: metal mesh or solid enclosures protect sensitive electronics from external electromagnetic interference. The metal body of a car also acts as a Faraday cage — passengers in a car struck by lightning are relatively safe because the charge flows around the outside of the conducting shell.

Particle Accelerators

All particle accelerators use electric fields to give charged particles kinetic energy. Linacs (linear accelerators) accelerate particles through a series of potential differences. Cyclotrons and synchrotrons use both electric and magnetic fields — the electric field increases the particles' speed, while the magnetic field bends their path in a circle. The Large Hadron Collider (LHC) accelerates protons to $6.5 \text{ TeV} = 6.5 \times 10^{12} \text{ eV}$ by passing them through millions of volts of potential difference many times around a 27-km ring.

Capacitors

A capacitor is a device that stores energy in an electric field. The simplest is the parallel-plate capacitor: two conducting plates separated by an insulator. The stored energy is $U = \frac{1}{2}CV^2$, where C is the capacitance (a geometric property, Chapter 42). Capacitors in flash units store the energy needed for a bright brief light pulse. Supercapacitors

(electrolytic double-layer capacitors) store enough energy for regenerative braking in hybrid vehicles.

41.14 Common Mistakes in Electrostatics Analysis

- Treating electric potential as a vector. Potential is a scalar. To find the potential at a point due to multiple charges, simply add the values algebraically (with signs). This is much simpler than adding electric field vectors, and students sometimes incorrectly try to add potentials as vectors.
- Confusing potential and potential energy. Potential $V = U/q$ is energy per charge; it is a property of the location in the field. Potential energy $U = qV$ is the energy of a specific charge q at that location. The same point in space has one value of V but different values of U depending on the charge placed there.
- Getting the sign wrong on work. $W = q\Delta V = q(V_B - V_A)$. A positive charge moving from low to high potential gains potential energy (work done against the field is positive). A positive charge moving from high to low potential loses potential energy and gains kinetic energy (the field does positive work). Negative charges behave oppositely.
- Forgetting that equipotentials and field lines are perpendicular. When sketching field line diagrams, many students draw field lines and equipotentials at arbitrary angles. They must be perpendicular everywhere.
- Using $E = V/d$ for non-uniform fields. $E = V/d$ applies only to uniform fields (parallel-plate geometry). For point charges and other geometries, $E = kQ/r^2$ and the field varies with position.
- Ignoring the sign of charges when computing potential. $V = kQ/r$ requires the sign of Q . A $+3\ \mu\text{C}$ charge gives positive V at all surrounding points; a $-3\ \mu\text{C}$ charge gives negative V . When superimposing, keep these signs: $V_{\text{total}} = V_1 + V_2 = k(+3\mu\text{C})/r_1 + k(-3\mu\text{C})/r_2$.
- Confusing the two uses of $E = V/d$ and $E = F/q$. $E = F/q$ defines the field in terms of force on a test charge. $E = V/d$ computes the magnitude of a uniform field from the potential difference and plate spacing. Both give E in V/m , but they apply in different situations.

Common Mistake: The most common electrostatics error: adding electric field vectors when potentials should be added, or vice versa. Use fields (vectors) when finding forces; use potentials (scalars) when working with energy or when superimposing contributions from multiple charges.

Chapter Summary

Electric fields and potentials describe how charges alter the space around them and how that alteration exerts forces and stores energy.

- Electric field $E = F/q$; SI unit $N/C = V/m$; vector pointing in direction of force on positive charge. Field of point charge: $E = kQ/r^2$.
 - Field lines: tangent to E ; away from + charges; toward – charges; never cross; density $\propto$ field strength.
 - Superposition: $E_{\text{total}} = \Sigma E_i$ (vector sum); $V_{\text{total}} = \Sigma V_i$ (scalar sum).
 - Force on a charge in a field: $F = qE$. Acceleration: $a = qE/m$.
 - Uniform field between parallel plates: $E = V/d$, directed from + to – plate.
 - Electric potential energy: $U = kQq/r$ (reference $U = 0$ at $r = \infty$).
 - Electric potential: $V = U/q = kQ/r$; SI unit volt ($V = J/C$); scalar.
 - Work moving charge through potential difference: $W = q\Delta V$.
 - Equipotential surfaces: $V = \text{constant}$; always perpendicular to field lines; conductors are equipotentials in electrostatics.
 - Field-potential relation: $E = -\Delta V/\Delta x$; field points from high to low potential.
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Key Terms

electric field (E): $E = F/q$; vector quantity giving the electric force per unit positive charge at each point in space; SI unit N/C or V/m.

field lines: Visual representations of the electric field; tangent to E , away from positive charges, toward negative charges, never crossing.

superposition principle: The total electric field (or potential) at a point is the vector (or scalar) sum of contributions from all individual source charges.

uniform electric field: A field of constant magnitude and direction; produced by parallel plates; field lines are parallel and equally spaced.

electric potential energy (U): $U = kQq/r$ for two point charges; energy stored in the configuration of charges; reference $U = 0$ at infinite separation; SI unit joule.

electric potential (V): $V = U/q = kQ/r$; potential energy per unit charge; scalar; SI unit volt ($V = J/C$).

potential difference (ΔV): $\Delta V = V_B - V_A = W/q$; the work per unit charge needed to move a positive charge from A to B; measured by a voltmeter.

electron-volt (eV): Unit of energy: $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$; the energy gained by one elementary charge accelerated through 1 V.

equipotential surface: A surface on which the electric potential is constant; always perpendicular to field lines; conductors are equipotential in static equilibrium.

Faraday cage: A conducting enclosure that shields its interior from external electric fields; interior $E = 0$ in electrostatic equilibrium.

parallel-plate capacitor: Two parallel conducting plates carrying equal and opposite charges; produces a nearly uniform field $E = V/d$ between the plates.

volt (V): SI unit of electric potential and potential difference; $1 \text{ V} = 1 \text{ J/C}$.

Concept Review Questions

Answer in complete sentences. No calculations unless stated.

1. Explain why physicists prefer the field model over direct action-at-a-distance for describing electric forces. What physical advantage does the field concept provide?
 2. The electric field at a point is directed to the north. A proton is placed at that point. What is the direction of the force on the proton? What if an electron is placed there instead?
 3. Explain the rules for drawing electric field lines. Why can field lines never cross?
 4. Two equal positive charges are placed 0.40 m apart. At the midpoint between them, what is the electric field? What is the electric potential? Explain why these are different (one is zero, the other is not).
 5. Explain what an equipotential surface is. Why are equipotentials always perpendicular to field lines? Use an analogy from topography if helpful.
 6. The electric potential inside a hollow metal sphere with charge on its outer surface is uniform and equal to the surface potential. What does this tell you about the electric field inside the sphere?
 7. Distinguish electric potential energy from electric potential. Which is a property of a specific charge at a location, and which is a property of the location itself?
 8. A positive charge is released from rest near another positive charge. Describe what happens to its kinetic energy, its potential energy, and the total mechanical energy as it moves away.
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Electric Field Problems

9. Find the electric field at a point 0.45 m from a charge of $-8.0 \mu\text{C}$. State the magnitude and direction.
 10. An electric field of $6.0 \times 10^4 \text{ N/C}$ acts on a charge of $-3.5 \mu\text{C}$. Find the force (magnitude and direction relative to the field).
 11. Two charges are on the x-axis: $+4.0 \mu\text{C}$ at $x = 0$ and $-4.0 \mu\text{C}$ at $x = 0.60 \text{ m}$. Find the electric field at the point $x = 0.30 \text{ m}$ (midpoint). (Hint: both fields point in the same direction at the midpoint for a dipole.)
 12. Find the point on the x-axis (other than infinity) where the electric field is zero, given $q_1 = +9.0 \mu\text{C}$ at $x = 0$ and $q_2 = +1.0 \mu\text{C}$ at $x = 1.0 \text{ m}$.
 13. An electron enters a uniform electric field $E = 3.0 \times 10^4 \text{ N/C}$ directed upward. The electron's initial velocity is $2.0 \times 10^6 \text{ m/s}$ horizontal. (a) Find the acceleration. (b) After the electron travels 0.050 m horizontally, how far has it been deflected vertically? (Use kinematics.)
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Potential Difference Exercises

14. Find the electric potential at a point 0.20 m from a $+5.0 \mu\text{C}$ charge and at a point 0.40 m from the same charge. Find the potential difference between these two points.
 15. Three charges are at the vertices of an equilateral triangle of side 0.30 m: $+2.0 \mu\text{C}$, $-2.0 \mu\text{C}$, and $+3.0 \mu\text{C}$. Find the electric potential at the centroid of the triangle. (Distance from centroid to each vertex = $0.30/\sqrt{3} \text{ m}$.)
 16. An alpha particle (charge $+2e$, mass $6.64 \times 10^{-27} \text{ kg}$) is accelerated from rest through a potential difference of 20 000 V. Find its final kinetic energy and speed.
 17. A charge of $+3.0 \mu\text{C}$ is moved from a point where $V = 200 \text{ V}$ to a point where $V = 500 \text{ V}$. (a) How much work must an external agent do? (b) Does the charge's potential energy increase or decrease? By how much?
 18. Parallel plates are 8.0 mm apart with a potential difference of 240 V. (a) Find the electric field between them. (b) Find the force on a proton between the plates. (c) Find the acceleration of the proton.
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Field Line Interpretation

19. Sketch the electric field line pattern for: (a) a single positive charge, (b) a single negative charge, (c) two equal positive charges side by side, (d) one positive and one negative charge of equal magnitude. For each, indicate the direction of the field at 4–6 representative points.
20. The field line density (number of lines per unit area) near charge A is four times greater than near charge B. What is the ratio of charge magnitudes $|q_A|/|q_B|$?
21. An electric field map shows equipotential lines at $V = 100\text{ V}$, 200 V , 300 V , and 400 V . Near one region, the equipotential lines are 1.0 cm apart; in another region they are 5.0 cm apart. Find the approximate electric field magnitude in each region. In which region is the field stronger?
22. A metallic sphere carries a large positive charge. Sketch the field lines and equipotential surfaces inside and outside the sphere. What is the field inside the sphere in electrostatic equilibrium?

Applied Electrostatics Problems

23. The deflection system of a cathode ray tube has plates 4.0 cm long separated by 1.2 cm . The potential difference between the plates is 180 V . An electron enters midway between the plates with horizontal speed $3.0 \times 10^7\text{ m/s}$. (a) Find E between the plates. (b) Find the vertical acceleration. (c) Find the vertical deflection of the electron while it is between the plates. (d) Is the approximation of treating the electron's path as parabolic reasonable here?
24. Two conducting spheres (radii 5.0 cm and 15 cm) are connected by a long thin wire. A total charge of $80\text{ }\mu\text{C}$ is placed on the system. (a) The charge distributes so that both spheres reach the same potential. Using $V = kQ/r$, find the charge on each sphere. (Hint: $V = kQ_A/r_A = kQ_B/r_B$ and $Q_A + Q_B = 80\text{ }\mu\text{C}$.) (b) Find the surface charge density ($\sigma = Q/4\pi r^2$) on each. Why is the surface charge density higher on the smaller sphere?
25. A proton and an alpha particle (α , charge $+2e$, mass $4u$) both start from rest and are accelerated through the same potential difference ΔV . Find the ratio of their final speeds v_α/v_p . (Use $KE = q\Delta V$ and $KE = \frac{1}{2}mv^2$.)
26. The electric potential in a region varies as $V = 4x^2 + 3y$ (in volts, with x and y in metres). (a) Find V at the point $(2.0, 1.0)$. (b) Estimate $E_x = -\Delta V/\Delta x$ at $x = 2.0\text{ m}$ by comparing V at $x = 2.0$ and $x = 2.1$ (holding y constant). (c) Estimate $E_y = -\Delta V/\Delta y$ at $y = 1.0\text{ m}$. (d) Find the magnitude of E at $(2.0, 1.0)$.

Chapter 41 Checkpoint

Before moving on to Chapter 42, confirm you can do each of the following.

I can...	See Section
Explain why physicists use the field model rather than action-at-a-distance	41.1
Define electric field as force per unit positive charge and give its SI unit	41.2
State the rules for drawing electric field lines	41.3
Calculate E at a distance r from a point charge	41.4
Apply the superposition principle to find E due to multiple charges	41.5
Find the force and acceleration of a charge in a known electric field	41.6
Describe the uniform field between parallel plates and use $E = V/d$	41.7
Calculate electric potential energy between two point charges	41.8
Calculate electric potential (voltage) due to a point charge	41.9
Find work done moving a charge through a potential difference	41.10
Describe and sketch equipotential surfaces; explain their relation to field lines	41.11
Use $E = -\Delta V/\Delta x$ to find field from potential or potential from field	41.12
Identify and avoid common errors in field and potential problems	41.14

Continue to Chapter 42: Capacitance and Capacitors →